

Electrodynamics

Notes

1 Recap/Intro

2 Today's stuff

Quantum Harmonic Oscillator.

We start given that some $\hat{H}\phi = E_n\phi$. From this, we know that:

$$\hbar\omega(\hat{a}_+\hat{a}_- + \frac{1}{2})\phi = E_n\phi$$

By the raising operator,

$$\begin{aligned}\hbar\omega(a_+a_-a_+ + \frac{1}{2}a_+)\phi &= E_n(a_+\phi) \\ \hbar\omega a_+(a_-a_+ + \frac{1}{2})\phi &= a_+ \left[\hbar\omega(a_-a_+ + \frac{1}{2}) \right] \phi \\ \hbar\omega(a_-a_+ - \frac{1}{2}) &= H \\ a_+(\hat{H} + \hbar\omega)\phi &= a_+(E + \hbar\omega)\phi = (E + \hbar\omega)a_+\phi\end{aligned}$$

Thus,

$$H(a_+\phi) = (E + \hbar\omega)(a_+\phi)$$

from which we can find that:

$$E_n = \hbar\omega \left(\frac{1}{2} + n \right)$$

Further, we MUST REMEMBER THAT:

$$\hat{a}_+\phi_n = \sqrt{n+1}\phi_{n+1}$$

$$\hat{a}_-\phi_n = \sqrt{n}\phi_{n-1}$$

$$\hat{a}_-\hat{a}_+\phi_n = n\phi_n$$

What do overlap integrals calculate? Know how to calculate them and find out what you are trying to match.